

# Two very different ways to spend 10 billion parameters

Qwen3.5-9B (dense, hybrid linear attention) vs GLM-4.7-Flash (31B-A3B MoE, MLA, MTP)

Intelligence + serving performance, measured on 1 H100 with Qwen3.5-27B as intelligence reference

80 serving points | 3 model eval suite with anchor reconciliation | FP8 ablation with quality gate  
every claim adversarially audited against raw data

# Four findings that decide deployments

**$c \approx 3$**

The dense/MoE crossover: GLM wins single-stream (+26%); Qwen wins everywhere past  $c = 4$ , peaking +84% at  $c = 32$ .

**4.1x**

more resident long-generation concurrency for the 9B: GLM's 63GB of weights crowd out its KV pool (measured 10 vs 42 requests at 49K).

**94%**

of the probe-derived bandwidth ceiling: the 9B's decode is roofline saturated at bs1 (2.0 TB/s measured copy, ~2.3 TB/s implied peak, not the SKU spec).

**+20 pts**

GPQA swing from token budget alone. Truncation corrected scores hit vendor anchors on all three models.

**Deployment summarized:** interactive and long context fleets take the 9B (or its FP8: +37% single-stream, no detected quality cost), math heavy low-QPS work takes GLM, with concurrency capped to its KV pool. Budget 2-4x the model card max-tokens for any reasoning workload.

**Provenance:** numbers extracted programmatically from raw result files (results/consolidated\_v2.json) with two adversarial audit passes that ran against the raw data, and what they caught (slide 13).

## The contenders: opposite systems bets

|                | Qwen3.5-9B                                                 | Qwen3.5-27B            | GLM-4.7-Flash                                             |
|----------------|------------------------------------------------------------|------------------------|-----------------------------------------------------------|
| Type           | dense 9.6B                                                 | dense 27.8B            | MoE 31.2B total / ~3B active                              |
| Attention      | hybrid 3:1 - 24/32 Gated DeltaNet (constant state) + 8 GQA | same hybrid, 64 layers | MLA (compressed KV latent), 47 layers + 1 MTP draft layer |
| KV bytes/token | ~32 KB                                                     | ~64 KB                 | ~53 KB                                                    |
| BF16 weights   | 19.3 GB                                                    | 55.6 GB                | 62.9 GB                                                   |
| License        | Apache-2.0                                                 | Apache-2.0             | MIT                                                       |

**The bets.** Qwen bets on memory with linear attention layers keeping the per-sequence state constant, so context barely grows the footprint. GLM bets on compute with ~3B active parameters per token + a speculative draft layer, but all 31B must live in VRAM. This study finds where each bet pays, and where each one's hidden cost lands (for the GLM being the KV pool and Qwen being a pre allocated state floor).

# The Method

**1 Primary-source ground truth:** every architecture claim verified against HF configs, release notes, engine docs before any design decision.

**2 Review before spending:** review passes found 30+ harness bugs pre-run (two would have invalidated the headline charts), plus two full audits of the final claims against raw data.

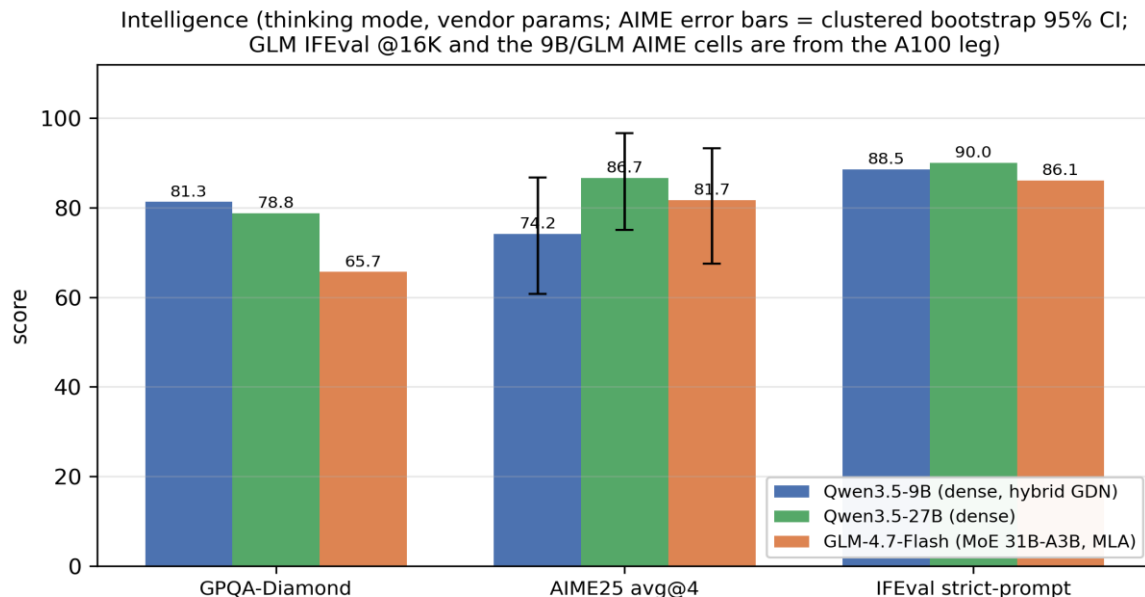
**3 Cheap validation:** the entire pipeline ran end to end on a \$0.43 spot instance before any serious GPU time.

**4 Guardrails in the harness:** warmup excluded: paired seeds for ablations, cache flushes, per-point /metrics forensics, p99 only at  $n \geq 300$  (enforced on charts), KV capacity logged per config, and in-situ bandwidth probe.

## What ran:

- 80 serving points: concurrency frontier, prefill/decode scaling to 64K, length variability, Poisson + burstiness + goodput, paired MTP ablation, prefix caching, FP8 weights + FP8 KV, KV-matched control
- GPQA-Diamond @32K, AIME25 multi-sample with bootstrap CIs, IFEval - all three models + FP8 quality gate
- Hardware: one H100 94GB (PCIe class bandwidth, driver 560), vLLM 0.22.1 via the cu129 wheel + CUDA minor version compat
- Disclosed gap: SGLang cross-check blocked by an sgl kernel torch ABI mismatch

# Intelligence: anchored, CI'd, and truncation accounted

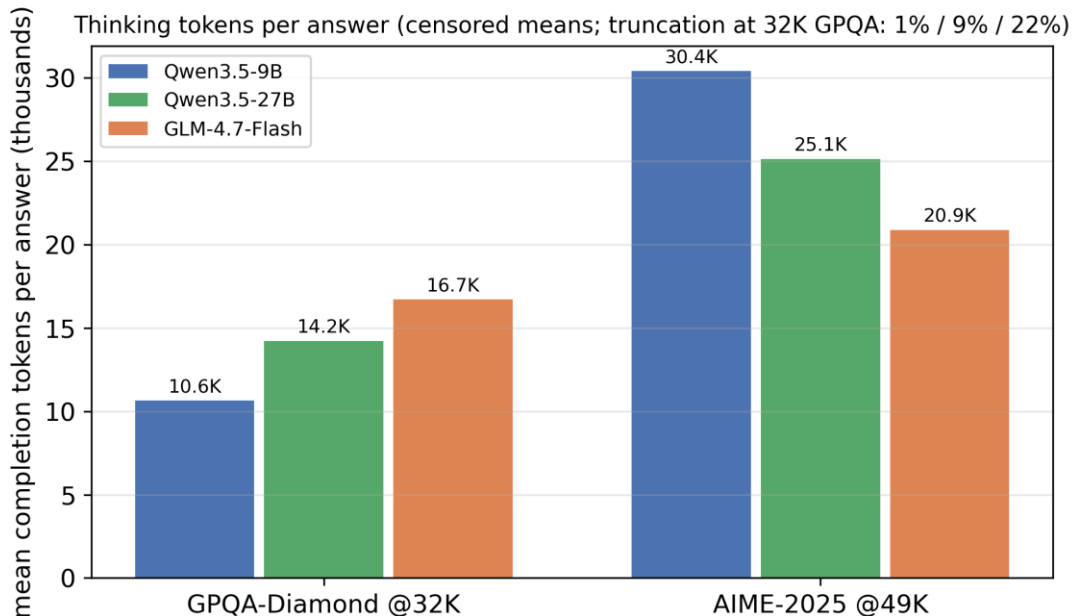


**Anchors hit three for three:** Truncation corrected GPQA lands on the vendor number for every model: 82.1 vs 81.7 (9B), 86.7 vs 85.5 (27B), bracket containing 75.2 (GLM). The harness earns trust the only way that counts (by reproducing known results).

**AIME with error bars:** 27B: 86.7  
GLM: 81.7 & 9B: 74.2. Clustered bootstrap CIs overlap; on 30 problems nobody should claim more than the intervals allow, so we don't.

*raw scores at 32K GPQA / 49K AIME budgets, thinking mode, vendor sampling params. Anchor comparison uses truncation-corrected GPQA (method in appendix); GLM IFEval and the 9B/GLM AIME cells are from the A100.*

# Eval scores are a function of token budget



45.5 → 65.7

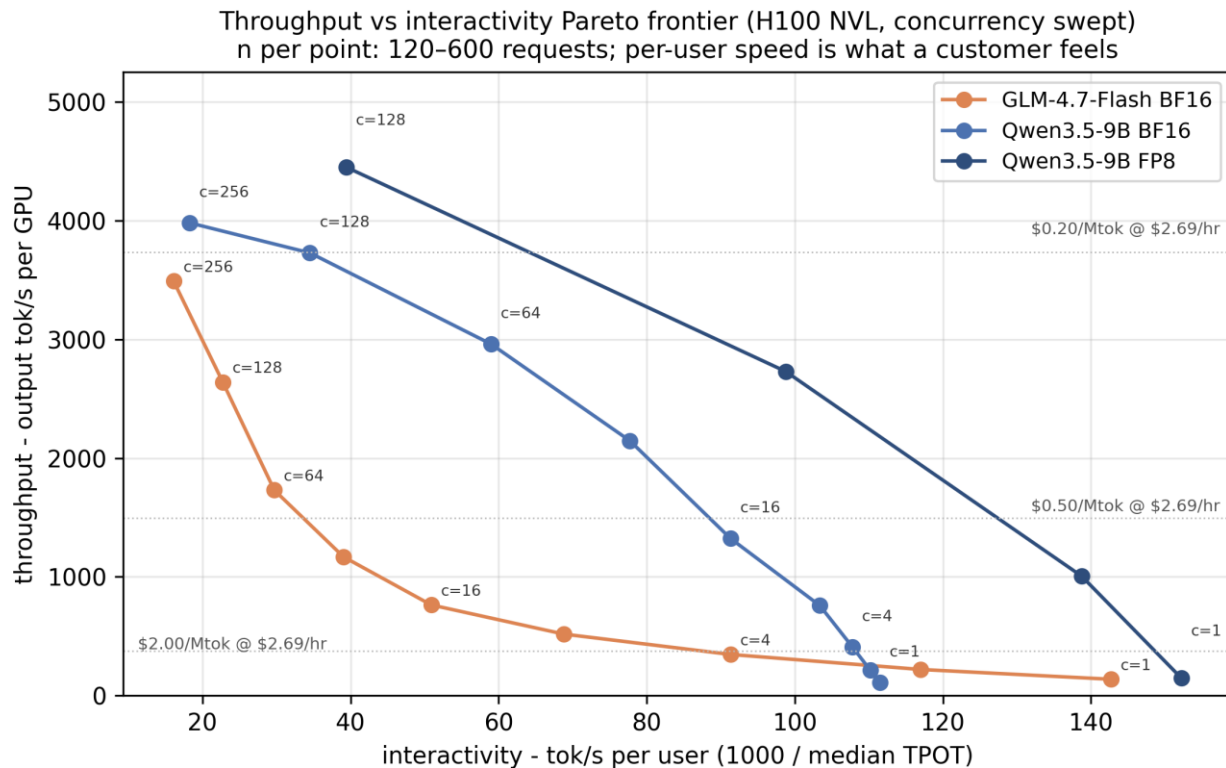
GLM GPQA at 16K vs 32K budgets: +20 points from budget alone, and 22% of answers still truncate at 32K

1% | 9% | 22%

truncation at the same 32K budget for 9B / 27B / GLM. Thinking length scales with capability and architecture. Budgets must too.

Mean tokens per answer: 9B - 10.6K, 27B - 14.2K, GLM - 16.7K on GPQA; 21-30K on AIME. Cost per answer and not cost per token, is the deployable metric, and which model is "cheaper" flips between different benchmarks

# Who wins where?

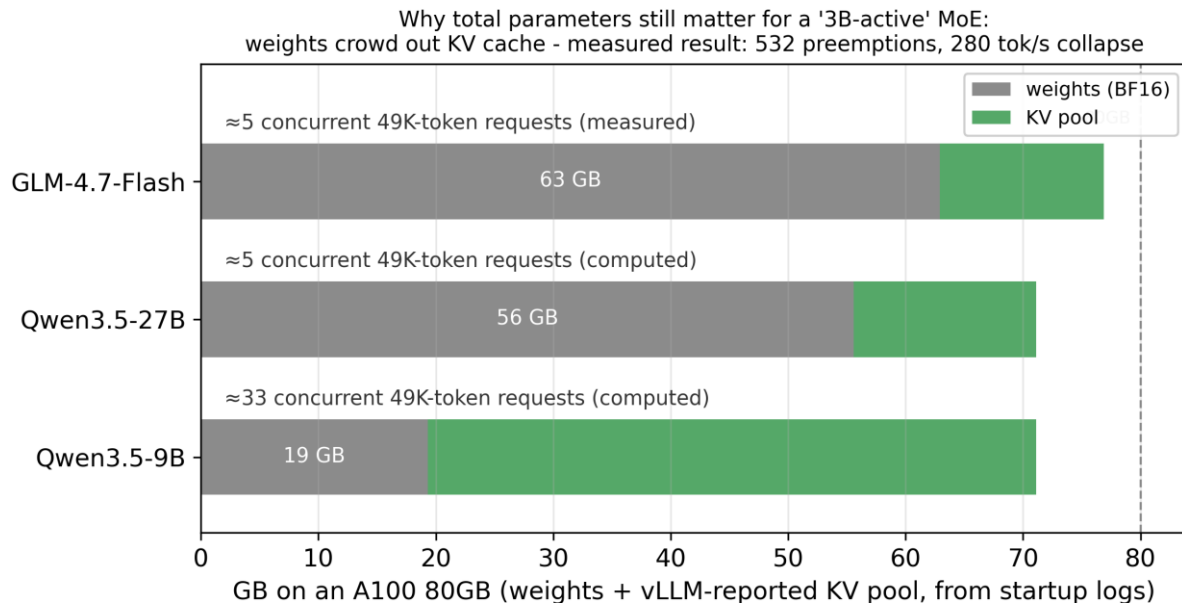


**Crossover at  $c \approx 3$ :** GLM's 3B-active FLOP bet wins single stream (137 vs 109 tok/s). Dense batching takes over by  $c = 4$  and peaks at +84% ( $c = 32$ ).

**FP8 dominates the middle:** Half the weight bytes pushes the whole frontier up and right. +37% single stream, decaying to +19% at  $c = 128$ .

**Endpoint:** At  $c = 256$  the gap narrows to +14% and GLM hasn't saturated. A re-cross beyond is plausible (untested).

# Batch × sequence length = KV memory



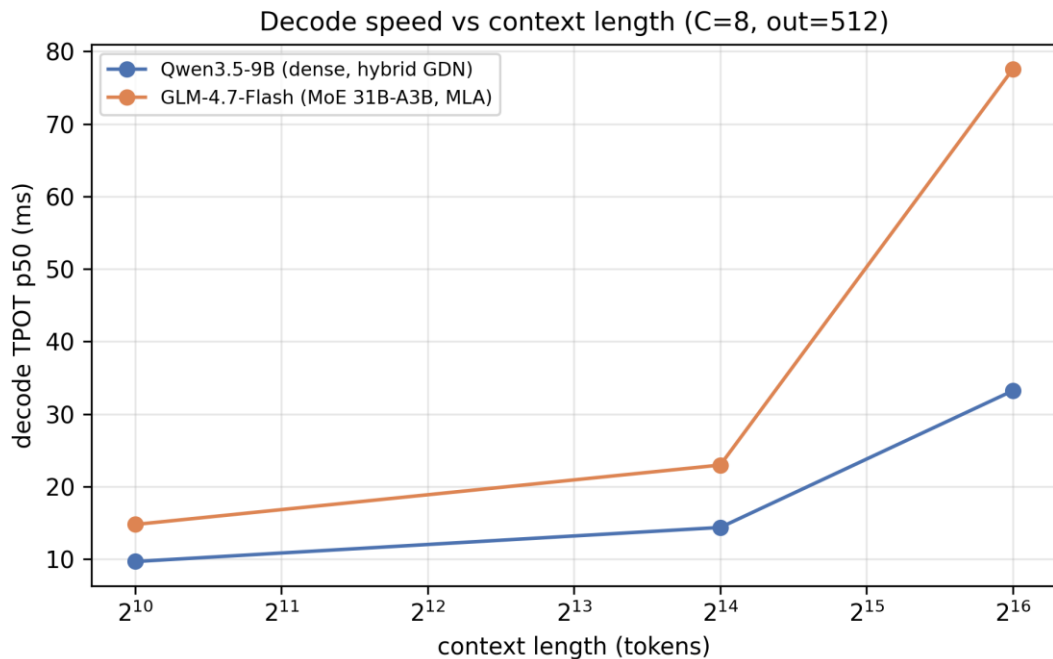
**Measured, both cards:** H100-94GB: 10 vs 42 resident 49K requests (4.1x). A100-80GB: 5 vs 33 (6x and Qwen-side pool from session logs, raw file). The ratio is card dependent, but the direction is not.

**The decomposition:** GLM's pool is about 4x smaller AND its per token KV is 1.7x higher than the hybrid's. MLA's reduction is real only against classical GQA. Against this competitor it's a deficit.

**Oversubscription episode (A100):** 532 preemptions, decode near 280 tok/s; consistent with thrash compounding slow long context decode (no clean baseline at that point and the H100 never preempted).



# Long context: architecture & not memory budget



## 2.3x

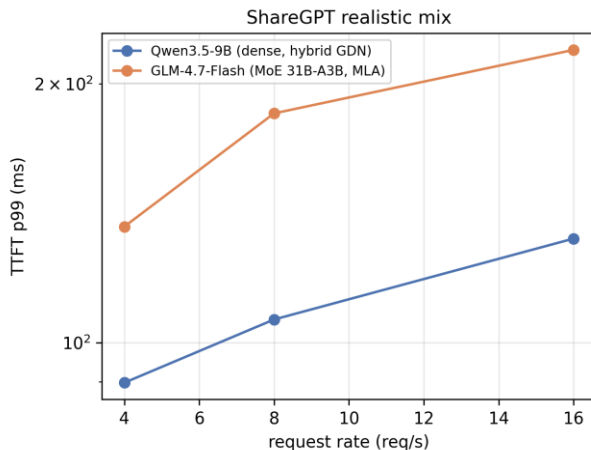
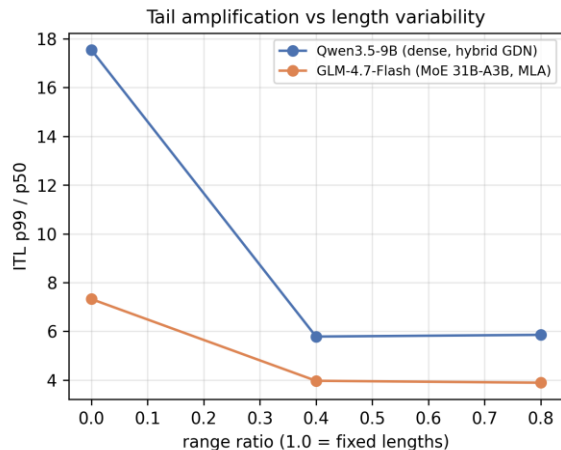
faster decode at 64K context (TPOT 33 vs 78 ms, matched concurrency, zero preemptions both sides). The constant state layers do exactly what they promise.

## 2.1x

the KV matched control: Qwen squeezed to a GLM sized memory budget still beats GLM at 64K (95.6 vs 46.0 tok/s). With equal memory and the same gap, we deduce it's the architecture.

The control crashed first with "max\_num\_seqs (1024) exceeds available Mamba cache blocks (907)". The hybrid's constant state is a pre-allocated per sequence floor. The fix (--max-num-seqs) and the re run are archived (server flags + the kvmatch point).

# Variable lengths and real traffic



**17.5x → 5.8x**

ITL tail ratio (p99/p50), fixed vs mixed lengths (9B; GLM 7.3x to 4.0x). Identical shapes synchronize completion waves. Throughput moves <4%.

**~2x**

lower median TPOT for the 9B on ShareGPT traces at every rate. 1.6x lower TTFT p99 and +22% at r = 16.

**r=8 → r=16**

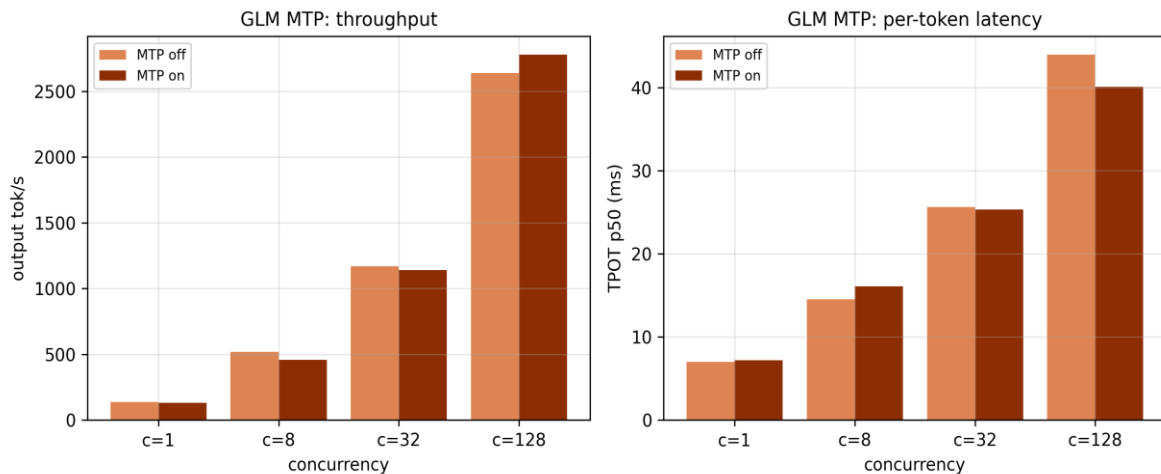
the goodput knee (TTFT < 1s, TPOT < 50ms): full goodput at r = 8, near zero by r = 16, both models. 2x burstier arrivals cost TTFT p99 only (+21% / +7%).

**+44% | 0%**

prefix caching, shared 2K prefix: GLM +44%; no-op on the 9B (vLLM auto disables it for hybrid-state models). FP8 KV on GLM: +1 to 5%. MLA is already compact.

Fixed shape benchmarks overstate inter token jitter and understate request time spread. Headline series in consolidated\_v2.json with full 80 point sweep in perf\_tidy.csv.

# Acceptance is the whole story: speculative decoding



Throughput counts emitted tokens only. Acceptance reported separately per point. Synthetic-prompt spec decode benchmarks structurally understate MTP. Ours measures both regimes.

Random-prompt sweep: -5.5 / -11.4 / -2.2 / +5.4% at c = 1 / 8 / 32 / 128 (single runs).

**-6% | +10%**

the same MTP flag on random token prompts (bs1) vs paired natural text (c = 8, identical prompts, natural EOS). Workload entropy decides the sign.

**51.3%**

draft acceptance on natural text, from the paired run's own counters. An earlier cumulative counter figure (29.6%) was caught by our audit.

# Roofline and energy, with a measured denominator

| config (bs1)            | measured  | ceiling*  | MBU                                |
|-------------------------|-----------|-----------|------------------------------------|
| Qwen3.5-9B BF16         | 111 tok/s | 119 tok/s | 94% - bandwidth saturated          |
| Qwen3.5-9B FP8          | 152 tok/s | 235 tok/s | 65%                                |
| GLM-4.7-Flash (S-MBU)   | 143 tok/s | 371 tok/s | 38-47% (active bytes basis)        |
| GLM total-weights basis | -         | -         | 390% = impossible. Proves sparsity |

\*weights-bytes / TPOT vs the ~2.3 TB/s implied peak derived from the in-situ copy probe (2.00 TB/s measured, large copies reach ~85-90% of peak). The PCIe class bandwidth, far from the NVL spec.

"Measured" here is the pure decode rate (1000 / mean TPOT). End to end output throughput runs lower (GLM bs1: 137 vs 143 tok/s) because it includes time to first token. The frontier slide uses the end to end number.

**2.02 → 0.19**

J/token from bs1 to c = 128 (GLM, GPU board power integrated over each run). Batching is a 10x energy lever before it's a cost lever!

## MoE - CAP

S-MBU accounting for the MoE (union of activated experts, not total weights). ML.ENERGY-consistent energy method; GPU-only, short-window rows flagged.

# Everything that broke

rsync silently ran absent on the control, a box was destroyed before local verify. Row level files of one run lost (headline numbers survived through live reads). **fix: salvage.sh which copies & verifies count + bytes, only then prints the destroy command. No error masking on decision bearing commands.**

Driver 560 vs CUDA13 wheels: forward compat rejected (NFB branch), six source builds hit nvcc 12.6 compiler crashes on FA3 kernels. **fix: CUDA minor-version compatibility - the cu129 release wheel runs on driver 560. Full diagnostic ladder is in CUDA\_COMPAT\_NOTES.md.**

KV-matched control crashed. The hybrid's GDN state pool is a pre-allocated floor (907 blocks < 1024 default seqs). Same mechanism killed the 27B eval server. **fix: --max-num-seqs capped per memory budget. The crash became finding #3.**

# What I'd run in production

Interactive chat / agent fleets

**Qwen3.5-9B (FP8)**

4x long-gen concurrency per GPU, +84% at typical operating points, bandwidth-saturated decode; FP8 with no quality drop detected (gate powered for ~7-pt effects).

Math quality at low QPS

**GLM-4.7-Flash**

Best single-stream speed; equal-or-better AIME at fewer thinking tokens; MTP +10% on natural text. Cap concurrency = KV pool / (context + thinking budget).

Any reasoning workload

**Budget for thinking**

2-4x the model-card max-tokens; alert on truncation rate and preemption count - both are silent quality and throughput killers.

Every recommendation traces to a measured artifact (frontier sweep, the KV-capacity logs, paired MTP point, the FP8 quality gate, the truncation ladder).

# What this study demonstrates

**Systems depth:** the findings live where architecture meets the scheduler, whether KV arithmetic with measured pools, a state-floor crash that became a finding, or an in-situ bandwidth probe that corrected the roofline.

**Statistical honesty:** CIs, truncation accounting, paired ablations, anchor reconciliation on all three models, and claims worded to exactly what single runs license.

**Operational maturity:** smoke before spend, budget guards, verified salvage, and a postmortem culture that turned every failure into either a fix or a finding.

**Reproducibility:** one command on any CUDA box with raw results, flags, versions, and analysis code all in the repo, and numbers extracted programmatically.